

Tailoring Machine Learning Weather Predictions for Local Impacts

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Running ML Weather Models is Cheap but Training them is Expensive

- Instead of spending compute power predicting upper atmosphere circulation patterns, why not focus on the weather variables most relevant to people
- Retraining and in some cases fully finetuning these models be out of reach for those poor in compute.

Goal: Demonstrate that is is possible to meaningfully improve the accuracy of relevant weather variables using a small amount of compute power and data.

Research Questions and Early Results

- Is it possible to improve forecast accuracy using easily available off the shelf tools?
- Can NWP and ML weather models can be corrected in the same way?
- We train a simple neural network to correct both a data driven ML forecast (Pangu) and a traditional NWP forecast (IFS HRES) we find success predicting 10m wind speed but not so much 2m temperature.
 - 10m Wind Speed Improvements range from 5%-20% for lead times 1 to 7 days
 - 2m Temp Improvements range from -20% - 20%

Key Contribution: Apply neural network post-processing to both NWP and ML weather models, showing that it is possible to improve the accuracy of both with perhaps some caveats.

Previous Papers have found Success Post-Processing NWP Forecasts with Neural Networks

- Han et al. 2021 trained a U-Net to correct IFS Forecasts in Northern China
 - Improvements in 2m Temp RMSE go range from 30% to 12% for lead times 1 to 7 days
 - Improvements in 10m Wind Speed RMSE Range from 25% to 20% for lead times 1 to 7 days
- Araujo and Andrade 2022 try to improve post-processing model GRIDPP over Norway
 - Improve RMSE of GRIDPP by about 1%. Suggests ML post-processing is most useful when there is not an existing and mature post-processing model.

Start by Manually Looking at Four Regions of Interest

- Choose a handful of regions with a variety of climates and land types.

`finetuning/climate_zones_with_patches.png`

Method

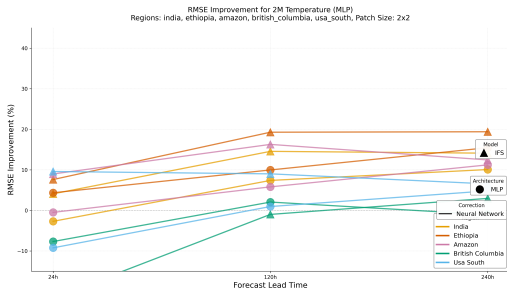
- Uses Pangu and IFS HRES forecasts with lead times from 1 to 7 days
 - Target Pangu predictions using ERA5 and IFS HRES predictions using its $t=0$ initialization
- Each model tries to predict a single variable for a single lead time in a particular region.
- Train neural network that takes in Pangu/IFS forecast outputs and targets MSE compared to target ERA5/IFS HRES data
 - Train using a single forecast per day for years 2018-2021
 - Test using daily forecasts in 2022
 - Test MLP architecture with 5 hidden layers, each with 512 neurons and a UNet architecture with 128 base channels

- Simple MLP method works across variety of regions and climates
- Training an MLP correction model for a 2x2 degree region takes about 2.5 minutes on an M3 max MacBook Pro
 - UNet model takes about 6 minutes to train.

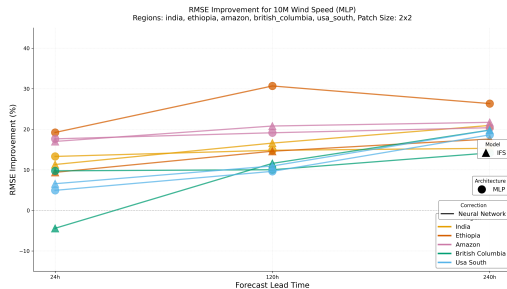
finetuning/pangu/lead_time/multi_region/2022/lead_time/lead_time_multi_region

Headline Result: Simple MLP Seems Effective at Correcting Wind Speed but not Temperature

- Simple MLP method works across variety of regions and climates
- The 2m temperature IFS predictions can be improved better than the Pangu predictions



(a) 2m Temperature: IFS vs Pangu

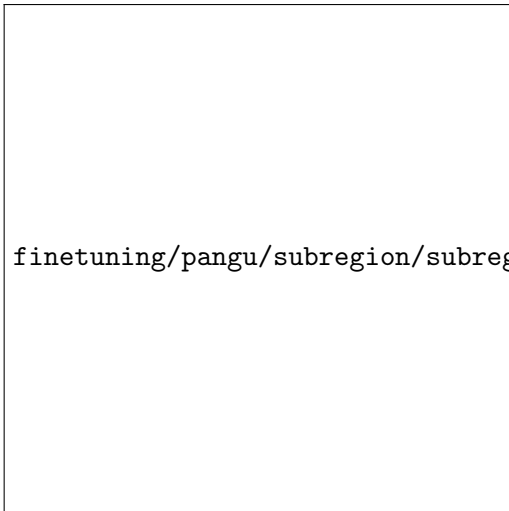


(b) Wind Speed: IFS vs Pangu

- Simple MLP method works across variety of regions and climates
- When it works, Neural net outperforms just subtracting the mean forecast error for each pixel

finetuning/pangu/lead_time/multi_region/2m2/lead_pangu/lead_time/multi_region

Effect Seems to mute with Longer Lead Times



(a) 2m Temperature 16°h: Pangu vs IES



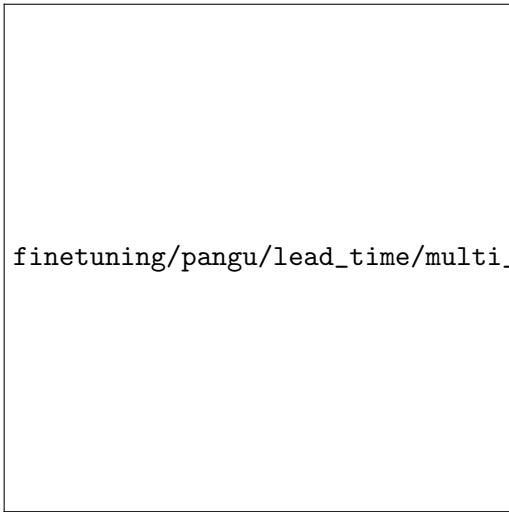
(b) Wind Speed 160h: Dangu vs IES

Exploring Heterogeneity by Climate Region

- Define tropical, temperate, and arid pixels on an 0.25 degree grid using broad Koppen-Geiger classes
- Train fine-tuning model on randomly sampled 2x2 degree patches

`finetuning/climate_zones_with_patches.png`

Heterogeneity by Climate Region



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu

Amazon Temperature

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finetuning/pangu/maps/amazon/2x2/rmse_maps_2m_temperature_trainedwith_2m_t
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Amazon Wind Speed

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finetuning/pangu/time_series/amazon/2x2/rmse_monthly_10m_wind_speed_train
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British Columbia Temperature

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British Columbia Wind Speed

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India Temperature

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India Wind Speed

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finetuning/pangu/time_series/india/2x2/rmse_monthly_10m_wind_speed_trained
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USA South West Temperature

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USA South West Wind Speed

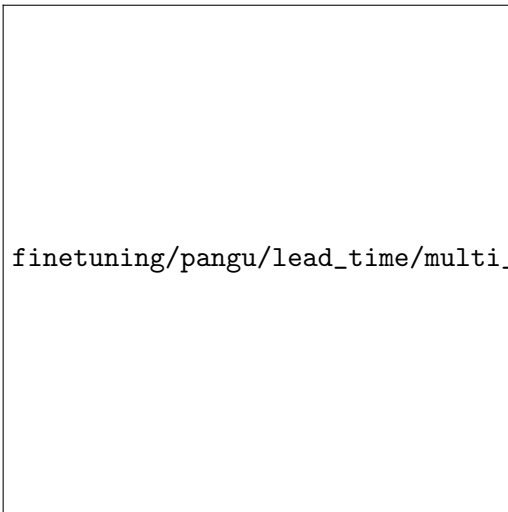
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What's next?

- Besides Pangu, what would be the most interesting ML weather model to try this on?
 - Interested in Gen/Graphcast and AIFS
- Use gridded observational data as training targets.
- Repeat same analysis using a daytime initialization time.

Forecast Improvement on 10x10 Degree Regions

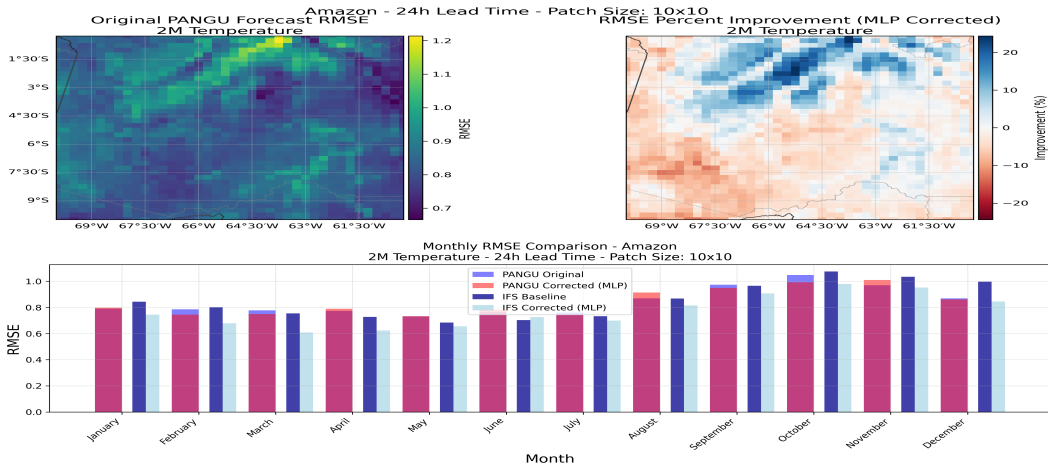


(a) 2m Temperature: IES vs Pangu



(b) Wind Speed: IES vs Pangu

Amazon Temperature 10x10



Amazon Wind Speed 10x10

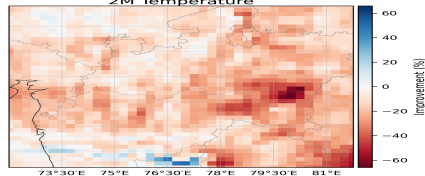
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India - 24h Lead Time - Patch Size: 10x10					
Original PANGU Forecast RMSE	Patch Size: 10x10	Improvement (%)	MLP Corrected RMSE	Percent Improvement (MLP Corrected)	% Error Reduction vs Original
2M Temperature	7.98	-27%	5.83	+24%	-48%



India Wind Speed 10x10

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finetuning/pangu/maps/india/10x10/rmse_maps_10m_wind_speed_trainedwith_10m
```

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finetuning/pangu/time_series/india/10x10/rmse_monthly_10m_wind_speed_train
```


USA South West Temperature 10x10

